

```

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report
import numpy as np

df = pd.read_csv("/Users/adilkhankaldybekov/Desktop/adil Khan/Machine Learning/Projects/Project for Uni 2/data/EU-Park-Customers.csv")
df['Pass_Type'] = df['Pass_Type'].fillna('None')

df = df.drop(columns=['Telephone_Number'])

y = df['Pass_Type']

X = df.drop(columns=['Pass_Type'])

le = LabelEncoder()
y_encoded = le.fit_transform(y)

X_train, X_test, y_train, y_test = train_test_split(X, y_encoded,
test_size=0.2, stratify=y_encoded, random_state=42)

model = LogisticRegression(multi_class='multinomial', solver='lbfgs',
max_iter=1000)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)
print(classification_report(y_test, y_pred, target_names=le.classes_))

/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/
site-packages/sklearn/linear_model/_logistic.py:1247: FutureWarning:
'multi_class' was deprecated in version 1.5 and will be removed in
1.7. From then on, it will always use 'multinomial'. Leave it to its
default value to avoid this warning.
  warnings.warn(

```

	precision	recall	f1-score	support
Gold	0.87	0.86	0.86	559
None	0.91	0.91	0.91	874
Silver	0.72	0.74	0.73	567
accuracy			0.85	2000
macro avg	0.83	0.83	0.83	2000
weighted avg	0.85	0.85	0.85	2000

```

df.describe()

```

	Age	Family_Members_with_Passes
Previously_Owned_Passes \		
count	10000.000000	10000.000000
mean	43.539400	0.752500
std	14.911636	0.891809
min	18.000000	0.000000
25%	31.000000	0.000000
50%	43.000000	1.000000
75%	56.000000	1.000000
max	69.000000	3.000000

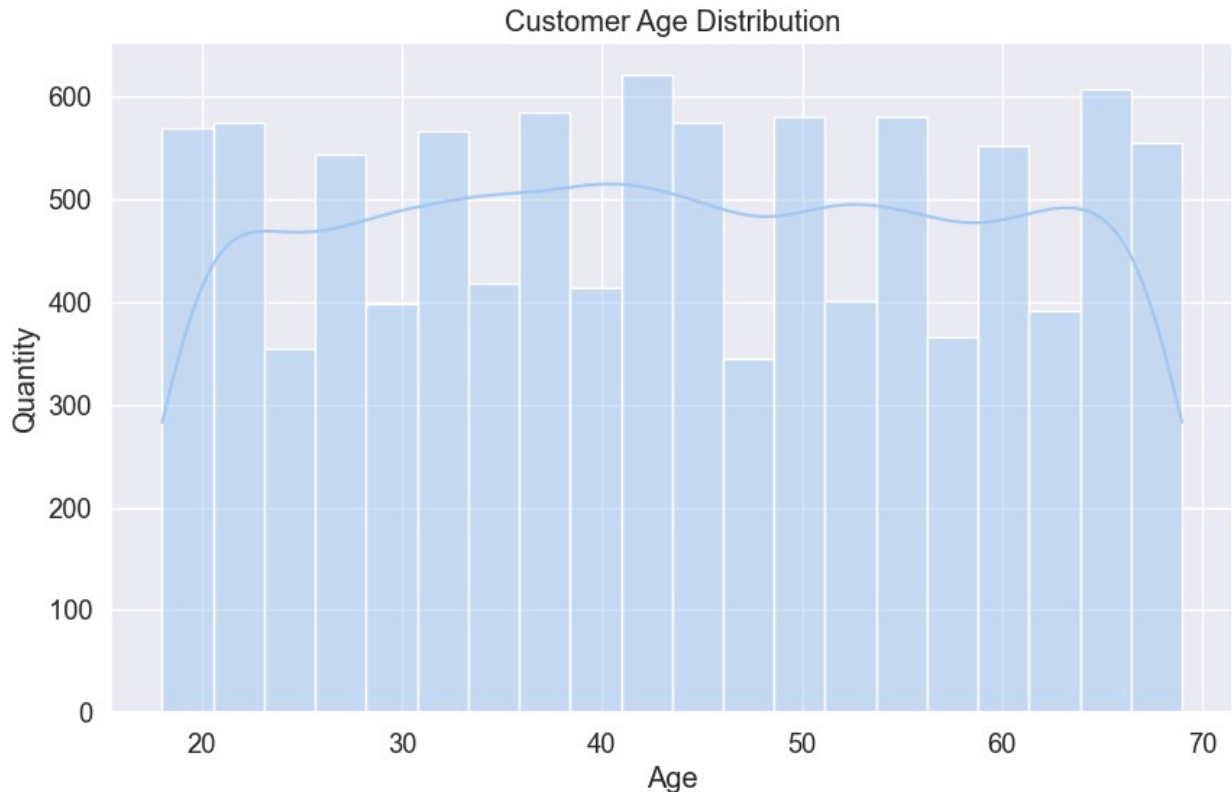
	Distance_from_Park_km
count	10000.000000
mean	29.936617
std	29.455316
min	0.000826
25%	8.600811
50%	20.788702
75%	41.993392
max	246.639786

```
sns.set_theme(style="darkgrid", palette="pastel", font_scale=1.2)
plt.rcParams["figure.figsize"] = (10, 6)
```

```
sns.countplot(x='Pass_Type', data=df,
order=df['Pass_Type'].value_counts().index)
plt.title("Pass_Type Classes Distribution")
plt.xlabel("Pass Type")
plt.ylabel("Quantity")
plt.show()
```



```
sns.histplot(data=df, x='Age', bins=20, kde=True)
plt.title("Customer Age Distribution")
plt.xlabel("Age")
plt.ylabel("Quantity")
plt.show()
```



```
#For binary classification of pass holders vs non-pass holders

df = pd.read_csv("/Users/adilkhankaldybekov/Desktop/adil Khan/Machine
Learning/Projects/Project for Uni 2/data/EU-Park-Customers.csv")
df['Pass_Type'] = df['Pass_Type'].fillna('None')
df = df.drop(columns=['Telephone_Number'])

df['Has_Pass'] = df['Pass_Type'].apply(lambda x: 1 if x in ['Gold',
'Silver'] else 0)

X = df.drop(columns=['Pass_Type', 'Has_Pass'])
y = df['Has_Pass']

X_train, X_test, y_train, y_test = train_test_split(X, y, stratify=y,
test_size=0.2, random_state=42)

model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)
print(classification_report(y_test, y_pred, target_names=["No Pass",
"Has Pass"]))

feature_importance = pd.DataFrame({
    'Feature': X.columns,
```

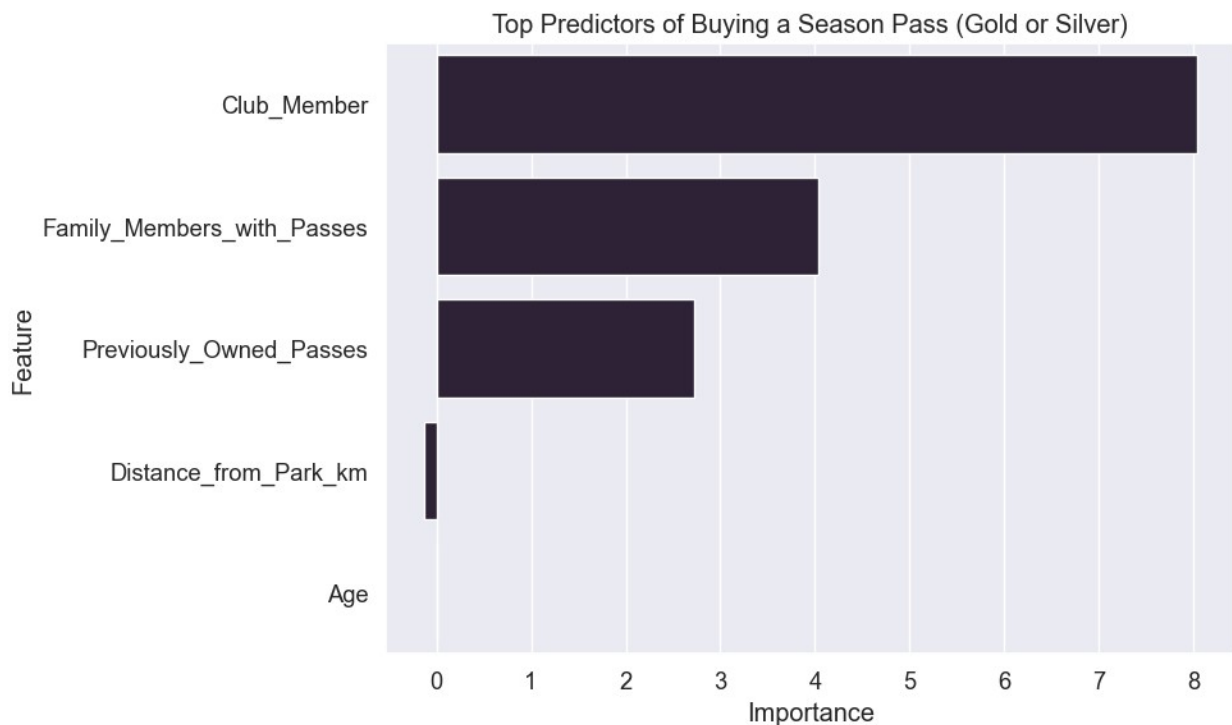
```

'Importance': model.coef_[0]
}).sort_values(by='Importance', key=np.abs, ascending=False)

sns.set_theme(style="darkgrid", palette="mako", font_scale=1.2)
plt.figure(figsize=(10, 6))
sns.barplot(data=feature_importance.head(10), x='Importance',
y='Feature')
plt.title("Top Predictors of Buying a Season Pass (Gold or Silver)")
plt.tight_layout()
plt.show()

```

	precision	recall	f1-score	support
No Pass	0.89	0.89	0.89	874
Has Pass	0.92	0.92	0.92	1126
accuracy			0.91	2000
macro avg	0.91	0.91	0.91	2000
weighted avg	0.91	0.91	0.91	2000

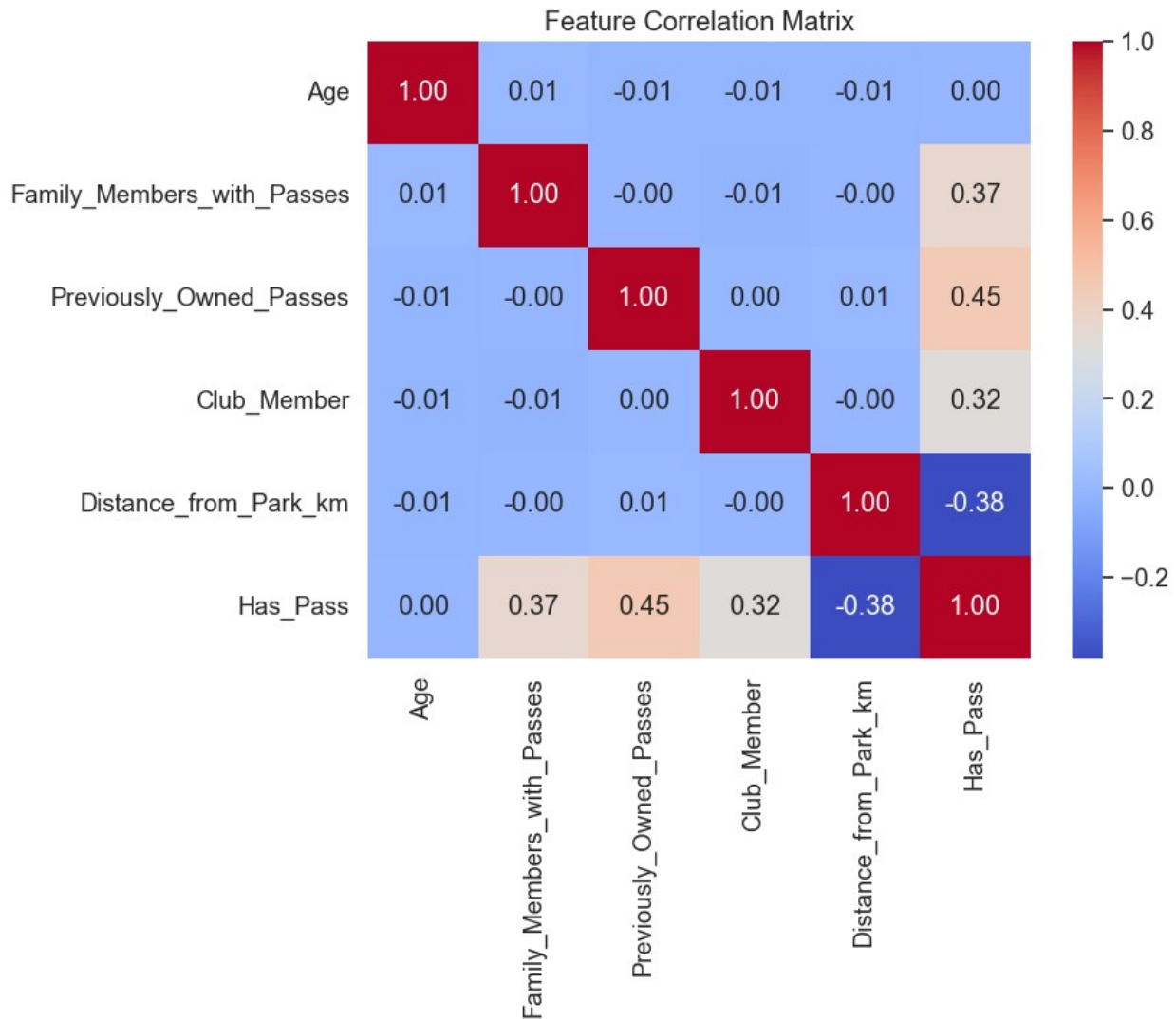


Interpretation: Top Predictors of Buying a Season Pass (Gold or Silver)

1. **Club_Member** – the strongest positive predictor
Customers who are club members are **significantly more likely** to purchase a season pass.
Insight: The membership program is highly effective and should be further promoted and expanded.

2. **Family_Members_with_Passes** – strong social influence
Having family members with passes **increases the likelihood** of a customer buying one too.
Insight: Social influence matters — referrals or family bundle offers could drive sales.
3. **Previously_Owned_Passes** – repeat behavior
Customers who have previously purchased passes are **more likely to buy again**.
Insight: This reflects customer loyalty. Targeted renewal offers could be very effective.
4. **Distance_from_Park_km** – mild negative impact
The farther a customer lives from the park, the **less likely** they are to buy a pass.
Insight: Consider local promotions or remote engagement options for distant customers.
5. **Age** – minimal effect
The customer's age appears to have **little influence** on the decision to purchase.
Insight: Age-specific marketing is likely not necessary.

```
corr = df.corr(numeric_only=True)
plt.figure(figsize=(8, 6))
sns.heatmap(corr, annot=True, cmap='coolwarm', fmt=".2f")
plt.title("Feature Correlation Matrix")
plt.show()
```



Final Recommendations for EU-Park Management (Task C)

Based on our analysis and predictive modeling, we recommend the following actions to increase Season Pass sales:

- Expand and Promote the Club Membership Program**
 Being a club member is the strongest predictor of pass purchase. EU-Park should incentivize customers to join the club (e.g., with discounts, exclusive benefits, or early access). Marketing should emphasize club benefits and target non-member visitors.
- Introduce Family Pass Bundles**
 Customers with family members who already have passes are significantly more likely to buy one. EU-Park should offer group/family pass discounts or "bring-a-family-member" campaigns to leverage this social effect.

3. **Target Repeat Buyers with Renewal Offers**

Previously owned passes are another strong signal of purchase intent. EU-Park should implement personalized renewal email campaigns and discounts for past season pass holders.

4. **Geographically-Tailored Promotions**

Distance from the park plays a complex role. LightGBM suggests that customers living within certain distance ranges are more likely to buy. EU-Park can use geo-targeted online ads, especially in nearby cities or regions with higher purchase probability.

5. **Avoid Over-Segmentation by Age**

Age has little predictive power. Instead of focusing on age groups, segment marketing by behaviors like club status or past purchases.